Inhalational Anesthetic

Authors

Amanda L. Miller[1]; Danny Theodore[2]; Jason Widrich[3].

Affiliations

[1] Chippenham Hospital
[2] UVA
[3] University of Florida

Last Update: May 8, 2022.

Continuing Education Activity

Inhalation anesthetics (nitrous oxide, halothane, isoflurane, desflurane, sevoflurane, most commonly used agents in practice today) are used for induction and maintenance of general anesthesia in the operating room. This review is a general overview of inhalation anesthetic agents. Inhalation anesthetic agents are medications primarily used in the operating room to provide general anesthesia for surgery. This activity describes the indications, action, and contraindications for the use of inhalational anesthetic agents. This activity will also highlight the mechanism of action, adverse event profile, and other key factors (e.g., off-label uses, dosing, pharmacodynamics, pharmacokinetics, monitoring, relevant interactions) pertinent for members of the interprofessional healthcare team in providing general anesthesia.

Objectives:

- Identify the mechanism of action of inhalation anesthetics.
- Describe the potential adverse effects of inhalational anesthetics.
- Review the appropriate monitoring for inhalational anesthetics.
- Outline some interprofessional team strategies for improving care coordination and communication to advance inhalational anesthetics and improve outcomes.

Access free multiple choice questions on this topic.

Indications

Inhalation anesthetics (nitrous oxide, halothane, isoflurane, desflurane, sevoflurane, most commonly used agents in practice today) are used for induction and maintenance of general anesthesia in the operating room. The volatile anesthetics (halothane, isoflurane, desflurane, and sevoflurane) are liquids at room temperature and require the use of vaporizers for inhalational administration. Nitrous Oxide is already under normal conditions of temperature and pressure. All inhalational anesthetics provide amnesia and immobility, except for nitrous oxide, which also provides analgesia. Inhaled anesthetics are commonly used in combination with IV anesthetic agents. These agents have FDA approval for use as a general anesthetic and sedation agent in the operating room. Inhaled anesthetic agents have also had use in the intensive care unit, but this is not an FDA-approved indication. The primary applications of inhaled anesthetic agents in the ICU are sedation, refractory bronchospasm, and control of status epilepticus unresponsive to anticonvulsant medications.

Mechanism of Action

The exact mechanism of action for inhaled anesthetics remains mostly unknown. Fundamentally, inhaled anesthetics work within the central nervous system by augmenting signals to chloride channels (GABA receptors) and potassium channels while depressing neurotransmission pathways. These pathways, including acetylcholine, both the muscarinic and nicotinic receptors, glutamate or NMDA receptors, and serotonin (5-HT receptors).[1] Inhalation agents are also sub-classified as either volatile and non-volatile.

Isoflurane, sevoflurane, and desflurane all decrease systemic blood pressure by decreasing systemic vascular resistance. For the most part, these agents preserve cardiac output, but cardiac depression can be seen if combined with other IV agents or in patients with acute cardiogenic shock. Desflurane has been known to cause hypertension and tachycardia with rapid administration of the agent.

Nitrous oxide can cause myocardial depression, but this effect is offset by the sympathetic increasing leading to minimal hemodynamic changes.

Volatile anesthetic agents are not true respiratory depressant drugs in the sense that they decrease the respiratory rate seen by other agents. They do decrease tidal volumes but with the respiratory rate increase. This is not equally matched; therefore, the minute ventilation can decrease.

Administration

The most commonly used anesthetic gases are halothane, nitrous oxide, isoflurane, sevoflurane, and desflurane. The primary mode of administration is by inhalation through a face mask, laryngeal mask airway, or a tracheal tube. They can be useful for preoperative sedation in addition to intravenous (IV) anesthetic agents such as midazolam and propofol in the perioperative and intraoperative setting.[2] The gold standard to measure potency is the minimum alveolar concentration (MAC), defined as the minimum alveolar concentration of inhaled anesthetic, at which 50% of people do not move in response to a noxious stimulus.[3] Each additional 0.1 above or below a MAC of 1.0 corresponds to a one standard deviation increase or decrease in dose. 50% of patients will not move at 1.0 MAC and 68% at 1.1 MAC, 95% at 1.2 MAC, and 99.7% at 1.3MAC.

Isoflurane has a MAC of 1.2% at sea level. MAC levels can change based on different factors. Old age, acute intoxication with sedating drugs, hypothermia will decrease the level of MAC. Hyperthermia, chronic cocaine abuse, and alcohol abuse will increase the level of MAC. After the age of 40, MAC levels decrease by 6% per decade.

Sevoflurane has a MAC of 2.6%. The age decrease in MAC value is the same as isoflurane.

Nitrous oxide, also known as "laughing gas," is a non-flammable, odorless, and colorless gas. It has a MAC of 104%. Unlike other inhaled anesthetics currently used, it has analgesic properties. Its use has fallen more into favor as an anesthetic option for laboring patients in the obstetric wards.

The higher the MAC, the lower the potency of gas is needed for sedation. Induction speed is determined by the alveolar concentration known as FA in conjunction with the inspired concentration known as FI. The rate at which this ratio approaches 1 is known as the speed of induction.[4] Achieving anesthetic goals also involves monitoring and directing the end-tidal CO2, tidal volume, and respiratory rate of the patient, often achieved when mechanically ventilated.

Children at 6 months of age will require a higher MAC, and the elderly often have reduced MAC requirements. Although 1.0 MAC is the number most often used to produce anesthesia and immobility, many practitioners also use a dose of anesthetic above a particular threshold to prevent recall called MAC awake. While not as reliable as the immobility measure, a dose above 0.4 to 0.5 MAC is considered sufficient to prevent recall. Recall under anesthesia is a rare event, especially with standard dosing of inhaled anesthetics. It has been reported and verified with patients who have had their memory tested post-procedure. Typically this occurs when lower doses of anesthetics were utilized.

Adverse Effects

The most common adverse effect of inhaled anesthetic agents is postoperative nausea and vomiting (PONV). There has been some evidence showing that intravenous anesthesia instead of inhaled agents reduces the risk of PONV.[5] Independent of the source, usually anti-emetic agents such as ondansetron, metoclopramide, and/or dexamethasone are administered both prophylactically and symptomatically to reduce the incidence of nausea and vomiting. [2]

Malignant hyperthermia (MH) is also an adverse effect that can occur with the administration of inhaled anesthetics, most commonly seen with the inhaled gas halothane. Patients susceptible to this adverse effect have heritable alterations between their proteins and muscular cytosolic concentrations of Ca2+.[6] When exposed to anesthetic gases, there is an excessive release of Ca2+ in the skeletal muscle causing the patient to exhibit symptoms such as hyperthermia, tachycardia, muscle rigidity, hyperkalemia, and metabolic imbalances. Reversal is achievable by administering dantrolene and restoration of normal body temperature, and correction of metabolic imbalances.[7] Patients with a known history or family history of MH should avoid volatile inhalation agents and other precipitating agents such as succinylcholine. Typically the volatile agent vaporizers are completely removed from the anesthesia machine, and it is flushed with high flow air or oxygen for an hour before being used with a susceptible patient.

A few inhalation agents are known to irritate the airways of patients with severe asthma and induce bronchospasm due to the pungent smell on induction, primarily with desflurane and isoflurane. Other agents like sevoflurane can be used in asthmatic patients to help relax the airways on induction as they do not have such pungent smells.

Isoflurane, sevoflurane, desflurane will decrease systemic vascular resistance leading to a drop in systemic blood pressure. These changes are more profound in hypovolemic patients.

Nitrous oxide can cause diffusion hypoxia quickly following discontinuation of the agent. It is recommended that 100% FiO2 be used to counteract the rapid dilution of O2 in the alveoli.

Contraindications

There are relatively few absolute contraindications. Most notably, individuals who have genetic contraindications, such as those that carry gene variations for malignant hyperthermia, should avoid anesthetic gases. Relative contraindications are patients with severe hypovolemia and those with severe intracranial hypertension as anesthetic gases might further decrease cerebral perfusion.

Nitrous oxide is contraindicated in patients undergoing craniotomies, bowel surgery, intraocular and middle ear surgeries. Nitrous oxide is thirty times more soluble than nitrogen. This leads to the rapid removal of nitrogen in these closed spaces. Nitrous oxide will quickly diffuse into these closed spaces leading to increased pressure/volume in the brain, bowel, eye, and middle ear. In addition, patients with pneumothorax or pulmonary hypertension can have worsening of the pneumothorax and increases in pulmonary hypertension related to the use of nitrous oxide.

Monitoring

The American Society of Anesthesiology (ASA) recommends that all patients undergoing general anesthesia with inhalation and/or IV anesthetic agents have standard ASA monitoring. Standard 1 involves the presence of qualified anesthesia personnel. Standard 2 requires monitoring of ventilation, oxygenation, temperature, and circulation. Ventilation is monitored with end-tidal CO2 (ETCO2) and inspired anesthetic gases, and oxygenation monitoring is with pulse oximetry (SpO2) and inspired O2 (with alarms). Temperature can be tracked via skin, esophageal, bladder rectal temperatures. Circulation monitoring is with continuous heart monitoring, blood pressure measurements every 5 minutes, and electrocardiogram.[8]

Intraoperative monitoring with a bispectral index (BIS), while not foolproof, can be helpful to assess for changes in the level of sedation. This EEG device will measure brain activity from 0 to 100, with values below 40 usually indicate deep sedation.[9] Minimum alveolar concentration (MAC) is also monitored in the operating room. Each inhalation anesthetic agent has a specific MAC. Nitrous oxide with a MAC value greater than 100% can not achieve acceptable levels for general anesthesia on its own. Halothane has a MAC value of 0.75%, isoflurane has a MAC value of 1.4%, desflurane has a MAC value of 6.6%, and sevoflurane has a MAC value of 2.0%.[2]

Inhaled anesthetics produce skeletal muscle relaxation and affect sensory nerve conduction. During procedures where motor evoked potentials and somatosensory evoked potentials are required for neuromonitoring, the recommendation is to use lower doses of all the inhaled anesthetics or eliminate them and add an intravenous anesthetic.

Toxicity

It is worth mentioning that there is no pharmacological intervention for an overdose of inhaled anesthetics. In an overdose incident, the primary treatment method is supportive, with optimal ventilator settings and alveolar clearance. Several rare acute and chronic toxicities can occur with inhaled agents. Acute toxicities include carbon monoxide poisoning (CO2), nephrotoxicity, and hepatotoxicity. Chronic toxicities include hematotoxicity, teratogenic effects, and carcinogenic toxicities.

The dose of Nitrous Oxide necessary used in a routine anesthetic can cause diffusion hypoxia. As gas exits the bloodstream into the lungs, the nitrous oxide displaces air and oxygen from the alveoli. This can be ameliorated by using supplemental oxygen to displace and dilute the nitrous oxide.

Hepatotoxicity and hepatic failure is rare but has associations with patients who have suffered exposure to halothane.[10]

Nephrotoxicity occurs most commonly with sevoflurane as its metabolism occurs at a much faster rate than other gases. This faster rate of absorption causes high levels of inorganic fluoride, which correlates with renal impairment. While this observation has largely occurred in research studies, the clinical recommendation remains to avoid sevoflurane in patients who have known renal dysfunction.[11][12]

Carbon monoxide toxicity with inhaled anesthetics occurs most commonly with desflurane as it is the largest producer of CO.[13] Inhaled anesthetics can produce additional CO as dry CO2 absorbers are used in the perioperative setting; CO accumulates when not changed.

Hematotoxicity is a chronic complication that can occur with prolonged exposure to nitrous oxide; this is due to a reduction in the recycling of vitamin B12.[14] If a patient has a chronic prolonged vitamin B12 deficiency, other symptoms such as megaloblastic anemia and neuropathies can begin to manifest. Additionally, when vitamin B12 metabolic pathways become disrupted, pregnant patients should be aware of potential teratogenic complications such as delayed cognitive impairment.[15]

Halothane produces a decrease in cardiac output with increasing dosage. Care should be taken using this anesthetic with patients known to have a low cardiac ejection fraction.

Neurologic and carcinogenic complications from volatile anesthetics or the choice of the anesthetic technique have not been reliably reproduced in human studies in randomized controlled studies.

Increasing awareness has been growing regarding chlorofluorocarbon waste from volatile anesthetic gases as a source of pollution. Their impact on the environment, and costs and benefits of volatile anesthetics, and their alternatives will continue to be a topic of debate for the foreseeable future.

Enhancing Healthcare Team Outcomes

An interprofessional team is the best design when administering inhaled anesthetics. A care team can consist of an anesthesiologist, certified registered nurse anesthetist, physician assistant, nurse practitioner, pharmacist, and other additional staff, including postoperative nurses and technicians. Volatile anesthetics are also used in veterinary medicine and administered by a veterinarian or a technician. Designating roles in this setting is essential. Having a devoted postoperative nurse who monitors vital signs is necessary as this is the primary means for monitoring a patient's safety and comfort. The pharmacy will play a role in storing and preparing these agents for administration and need to be aware of all medications the patient is taking. This type of interprofessional teamwork is crucial to achieving optimal results when using inhaled anesthesia. [Level 5]

The use of inhaled agents has also demonstrated benefits in the critical care setting, where they are often utilized in conjunction with intravenous anesthetics to provide optimal sedation for urgent procedural interventions. Their role outside of the operating room has increased as procedural suites outside the traditional operating room have also increased. Inhalational anesthetic agents have been used more in medically refractory epilepsy patients in neurointensive care units. These agents have also been used to treat status asthmaticus refractory to medical management. In this setting, physician assistants, nurse practitioners, critical care nurses, and ICU staff should have a fundamental knowledge of inhaled anesthetic agents. It is important to have essential care staff abreast of inhaled agents, and signs and symptoms of complications since this staff often see these patients in the postoperative anesthesia care unit or intensive care unit.